Software Requirements Specification (SRS)For FleetGuard – Smart Transport Management System

Version 2.0-----Table of Contents

-----1. Introduction1.1 Purpose

This Software Requirements Specification (SRS) document provides a comprehensive description of the requirements for **FleetGuard – Smart Transport Management System Version 2.0**. This document outlines all functional and non-functional requirements, system features, interfaces, and constraints for the enhanced system, which now includes customer registration, real-time location tracking, OTP verification, and multi-channel communication features.1.2 Document Conventions

- **Shall/Must:** Indicates a mandatory requirement that must be implemented.
- **Should:** Indicates a recommended requirement that is desirable but not mandatory.
- **May:** Indicates an optional requirement.
- Requirements are numbered hierarchically for easy reference (e.g., 3.2.1).

1.3 Intended Audience and Reading Suggestions

- **Project Managers:** To understand project scope and plan development phases (Sections 1, 2, 8).
- **Development Team:** To understand functional and technical requirements (Sections 3, 4, 5, 6).
- **Quality Assurance Team:** To design test cases (Sections 3, 4, 5, 7).
- **Stakeholders/Clients:** To review and approve system functionality (Sections 1, 2, 3, 10).

1.4 Product Scope

FleetGuard is a web-based transport management system that connects customers, drivers, and administrators on a single platform. The system enables customers to create shipments, drivers to accept assignments, real-time location tracking, secure OTP verification for delivery completion, and multi-channel communication between customers and drivers.

The software provides:

- Customer registration and shipment creation.
- Driver registration with license and experience details.
- Real-time automatic location tracking (every 30 minutes).
- OTP-based delivery verification.
- Accident and emergency reporting with fund requests.
- Multi-channel communication (calling, chat, phone number display).
- Admin oversight and management.

1.5 References

- Twilio API Documentation: https://www.twilio.com/docs (for SMS/Email OTP).
- WebRTC Documentation: https://webrtc.org/ (for internet calling).
- Socket.IO Documentation: https://socket.io/ (for real-time chat).
- Razorpay Payment Gateway API: https://razorpay.com/docs/.
- Google Maps/OpenStreetMap API: For location tracking.

-----2. General Description2.1 Product Perspective

FleetGuard is an enhanced, self-contained system with three user types: **Customer**, **Driver**, and **Admin**. The system operates as a web application with real-time features including automatic location tracking, instant notifications, OTP verification, and communication capabilities.

System Components:

- **Frontend:** Responsive web interface for customers, drivers, and admin.
- **Backend:** Application server handling business logic, WebSocket connections.
- **Database:** MySQL for persistent data storage.
- **Third-Party Services:** Twilio (SMS/Email OTP), WebRTC (calling), Razorpay (payments).
- **Location Service:** GPS/Geolocation API for automatic tracking.

2.2 Product Functions

Module	Key Functions
User Registration & Login	Register customers/drivers; email & phone verification via OTP.
Shipment Management	Create shipments; assign to drivers (self-assign or admin-assign).
Real-Time Location Tracking	Automatic location updates every 30 minutes; visible to admin and customers.
OTP Verification	Send OTP at shipment start; verify OTP at delivery completion.
Accident & Emergency	Report accidents (small/major/critical); fund requests; admin alerts.
Communication	Internet calling, chat window, phone number display.
Admin Dashboard	Monitor all shipments, drivers, customers, and emergencies.

Truck & Driver Management	Manage fleet and driver details.

2.3 User Characteristics2.3.1 Customer

- **Technical Expertise:** Basic web/mobile navigation skills.
- **Frequency of Use:** As needed for creating shipments.
- **Responsibilities:** Register, create shipments, track driver location, communicate with driver, verify OTP.
- **Access Level:** Customer-specific functions only.

2.3.2 Driver

- **Technical Expertise:** Moderate (GPS permissions, app navigation).
- **Frequency of Use:** Daily during trips.
- **Responsibilities:** Accept/reject shipments, share location, report emergencies, communicate with customers.
- **Access Level:** Driver-specific functions, location permission required.

2.3.3 Admin

- **Technical Expertise:** Moderate to high computer literacy.
- **Frequency of Use:** Daily during business hours.
- **Responsibilities:** Manage drivers and trucks, monitor shipments, respond to emergencies, assign shipments.
- **Access Level:** Full system access.

2.4 Operating Environment

- **Platform:** Web-based application accessible through modern browsers.
- **Mobile Compatibility:** Fully responsive for smartphones and tablets.
- **Server:** Cloud platform (AWS, GCP, or Azure) running Linux/Windows.
- **Database:** MySQL 5.7 or higher.
- **Backend:** Python (Flask/Django) or Node.js.
- **Real-Time Services:** WebSocket server for chat and notifications.
- **External APIs:** Twilio (SMS/Email), WebRTC (calling), Razorpay (payments).

2.5 Design and Implementation Constraints

- **Location Permission:** Driver must grant location permission for automatic tracking.
- **Internet Dependency:** Stable internet required for real-time features.
- **Browser Support:** Modern browsers with WebRTC support for calling.
- **OTP Validity:** OTP shall expire after 10 minutes.
- **Location Update Frequency:** Fixed at 30-minute intervals.

2.6 Assumptions and Dependencies

Assumptions:

- Drivers have smartphones with GPS capability.

- Customers have access to email and mobile phones.
- Internet connectivity is available to all users.
- Users consent to location sharing and communication features.

Dependencies:

- Twilio API for SMS and email OTP delivery.
- WebRTC-compatible browsers for internet calling.
- Geolocation API availability on driver devices.

-----3. Functional Requirements

3.1 User Registration and Login Module

3.1.1 Driver Registration

ID	Requirement Description	Priority
FR-101	New drivers shall register by providing: First Name, Surname, Username, Phone, Email, Password, License Number, Experience (in years).	High
FR-102	The system shall validate that username, phone, and email are unique.	High
FR-103	Upon registration, the system shall send OTP to the provided phone number via SMS and to the email address.	High
FR-104	The driver shall not be registered until both OTPs are verified.	High
FR-105	OTP shall expire within 10 minutes of sending.	High
FR-106	After successful verification, the driver account shall be activated.	High

3.1.2 Customer Registration

ID	Requirement Description	Priority
FR-201	New customers shall register by providing: First Name, Surname, Username, Phone, Email, Password.	High
FR-202	The system shall validate that username, phone, and email are unique.	High
FR-203	Upon registration, the system shall send OTP to the provided phone number via SMS and to the email address.	High

FR-204	The customer shall not be registered until both OTPs are verified.	High
FR-205	OTP shall expire within 10 minutes of sending.	High

3.1.3 Login Module

ID	Requirement Description	Priority
FR-301	Existing users (Customer, Driver, Admin) shall log in using Username and Password.	High
FR-302	The system shall authenticate users and redirect to role-specific dashboards.	High
FR-303	Upon unsuccessful login, appropriate error message shall be displayed.	High
FR-304	The system shall maintain user sessions with automatic timeout after inactivity.	High

3.2 Shipment Creation and Assignment Module

3.2.1 Shipment Creation (Customer)

ID	Requirement Description	Priority
FR-401	Customer shall be able to create a new shipment with details: Pickup Location, Delivery Location, Item Type, Weight, Dimensions (optional).	High
FR-402	The system shall automatically generate a unique Shipment ID.	High
FR-403	Upon creation, the shipment status shall be set to "Pending Assignment".	High
FR-404	The new shipment shall be visible in notification sections (header) to both admin and eligible drivers, who can self-assign themselves to the shipment.	High

3.2.2 Automated Intelligent Dispatch Engine (NEW)

ID	Requirement Description	Priority
FR-410	Dispatch Algorithm Trigger: Upon creation of a new shipment (FR-401), the system shall automatically trigger the Intelligent Dispatch Engine to evaluate the optimal driver-truck assignment.	High

FR-411	Multi-Parameter Matching: The Dispatch Engine shall calculate a compatibility score for each eligible driver-truck pair based on the following weighted parameters: Driver's real-time proximity to pickup location, Driver's current Hours-of-Service (HOS) remaining, Driver's historical on-time performance rating, Truck's cargo capacity vs. shipment weight/volume, Truck's equipment compatibility (e.g., refrigeration, liftgate), and Driver's certification match (e.g., HAZMAT).	High
FR-412	Configurable Optimization Goals: The Admin shall be able to configure the Dispatch Engine's primary optimization goal from the Admin Settings panel. Available modes shall include: MINIMIZE_DEADHEAD (prioritizes closest driver), BALANCE_WORKLOAD (distributes shipments evenly), and MINIMIZE_COST (prioritizes fuel-efficient truck-driver combinations).	Medium
FR-413	Automated Assignment with Confidence Threshold: If the Dispatch Engine identifies a driver-truck pair with a compatibility score exceeding the configurable threshold (default: 85%), the system shall automatically assign the shipment to that driver, update the shipment status to "Assigned", and send push notifications to both the Customer and Driver.	High
FR-414	Human-in-the-Loop Fallback: If no driver-truck pair meets the confidence threshold, the shipment shall be flagged as "Requires Manual Assignment" and appear prominently in the Admin Dashboard with a ranked list of the top three recommended driver-truck pairs for Admin review and final assignment.	High
FR-415	Assignment Audit Trail: The system shall log all dispatch decisions (both automated and manual) with timestamp, algorithm version, matching parameters used, compatibility score, and the identity of the Admin who performed any manual override. This log shall be accessible for operational review.	Medium

FR-416	Driver Notification & Acceptance Window: For automated assignments, the driver shall receive a push notification with shipment details and have a configurable window (default: 5 minutes) to either accept or reject the assignment. If rejected, the Dispatch Engine shall immediately re-evaluate the next driver without manual intervention.	High

3.2.3 Manual Assignment Override (Admin)

ID	Requirement Description	Priority
FR-407	Admin shall be able to manually assign a shipment to an available driver.	High
FR-408	The system shall update the shipment status to "Assigned" once a driver is confirmed.	High
FR-409	The system shall notify the customer and driver upon assignment.	High
FR-417	Manual Override Reason Capture: When an Admin manually assigns a shipment that differs from the Dispatch Engine's top recommendation, the system shall prompt the Admin to select a reason from a dropdown (e.g., "Driver Requested Load," "Customer Preference," "Equipment Mismatch"). This data shall be used to refine the algorithm's future recommendations.	Low

3.3 Real-Time Location Tracking Module

ID	Requirement Description	Priority
FR-501	The driver's device shall automatically push location data every 30 minutes.	High
FR-502	The system shall display the truck's real-time location on a map for the Admin user.	High
FR-503	The system shall display the truck's real-time location to the associated Customer.	High
FR-504	Location data shall be timestamped and stored in the database.	High
FR-505	The driver shall be able to temporarily pause automatic location sharing for a maximum of 1	High

	hour (Admin alerted).	
FR-506	If the driver has not updated location for 60 minutes, the system shall alert the Admin and Customer.	High

3.3.1 AI-Driven Route Optimization

ID	Requirement Description	Priority
FR-507	Initial Optimized Route Generation: Upon shipment status changing to "In Transit," the system shall automatically calculate an optimized route from pickup to delivery location using the AI Route Optimization Engine. The engine shall consider static constraints (vehicle height/weight restrictions, road classifications) and real-time data (current traffic conditions, weather forecasts).	High
FR-509	Driver Re-Route Notification: When a viable alternative route is identified (FR-508), the system shall send a push notification to the driver's device displaying: Current Delay, Proposed New Route ETA Savings, and Distance Difference. The notification shall include "Accept Re-Route" and "Ignore" buttons.	High
FR-510	ETA Update Propagation: Upon driver acceptance of a re-route, the system shall automatically recalculate the ETA and push the updated arrival time to the Customer's dashboard and the Admin's monitoring interface via WebSocket.	High
FR-511	Hours-of-Service (HOS) Aware Routing: The Route Optimization Engine shall integrate with the Driver's available HOS data. The system shall not propose a route that would require the driver to exceed legal driving hours without a scheduled rest stop. If the destination cannot be reached within HOS limits, the system shall proactively suggest an optimal rest stop location along the route.	High
FR-512	Route Deviation Detection: The system shall continuously compare the driver's actual GPS path against the optimized route. If a deviation exceeding 3 kilometers is detected without an accepted re-route or reported emergency, the system shall flag this as an "Unauthorized Route	Medium

	Deviation" alert on the Admin Dashboard for follow-up.	

3.3.2 Enhanced Map Visualization Requirements

ID	Requirement Description	Priority
FR-515	Route Overlay Visualization: The Customer and Admin map views shall display the driver's current optimized route as a colored line overlay on the map, with the planned path distinguished from the traveled path (e.g., Green = Traveled, Blue = Planned Route).	High
FR-516	Waypoint and Rest Stop Display: The map shall display icons for upcoming waypoints, scheduled rest stops (based on HOS calculations), and predicted traffic congestion zones along the route.	High

3.4 OTP Verification for Shipment Completion Module

ID	Requirement Description	Priority
FR-601	Upon shipment start, the system shall generate a unique 6-digit OTP and send it via SMS/Email to the Customer.	High
FR-602	The driver shall require the Customer to provide the OTP before marking the delivery as complete.	High
FR-603	The system shall only change shipment status to "Delivered" upon successful OTP verification.	High
FR-604	If OTP verification fails 3 times, the system shall notify the Admin.	High
FR-605	The Admin shall be able to manually override the OTP process only if verification fails 3 times.	High
FR-606	The system shall record the OTP, verification timestamp, and user who performed the verification.	High

3.5 Accident and Emergency Reporting Module

ID	Requirement Description	Priority
FR-701	The driver shall be able to report three levels of accidents: Small, Major, and Critical.	High

FR-702	For Major/Critical accidents, the system shall automatically freeze the shipment status and alert the Admin instantly.	High
FR-703	The driver shall be able to request emergency funds (Amount, Purpose) during reporting.	High
FR-704	The Admin dashboard shall display reports with color-coding based on severity (Critical = Red).	High
FR-705	The Admin shall be able to view, approve/reject fund requests, and update report status.	High
FR-706	The system shall automatically notify the Customer of a shipment delay due to an emergency.	High
FR-707	Breakdown reporting shall remain separated from accident reporting (using existing functional module structure).	High

3.6 Communication Module (Driver-Customer)

ID	Requirement Description	Priority
FR-801	The system shall provide an in-app chat window for direct text communication between the assigned Driver and Customer.	High
FR-802	The system shall allow the Driver to initiate an internet call (WebRTC) to the Customer directly through the app interface.	High
FR-803	The system shall allow the Customer to initiate an internet call (WebRTC) to the Driver directly through the app interface.	High
FR-804	The system shall allow the Driver's phone number to be displayed to the Customer only once the shipment status is "In Transit".	High
FR-805	All chat messages shall be stored and retrievable by Admin for auditing purposes.	High

3.7 Admin Dashboard Module

ID	Requirement Description	Priority
FR-901	The dashboard shall display current total counts for all Customer, Driver, and Truck records.	High

FR-902	The dashboard shall display a map view showing the location of all trucks marked "In Transit".	High
FR-903	The dashboard shall show pending approval requests for emergency funds.	High
FR-904	The dashboard shall display a list of all active shipments with their location update frequency status.	High
FR-905	The dashboard shall display alerts for drivers who have paused location sharing.	High
FR-906	The dashboard shall facilitate quick navigation to manage shipments, trucks, and users.	High
FR-907	**Predicted Delay Alert Feed:** The dashboard shall include a dedicated widget that displays a list of shipments flagged as "High Risk of Delay." The system shall calculate this risk based on real-time traffic data, weather conditions, and driver Hours-of-Service (HOS) status. The alert shall display the Shipment ID, Driver Name, and Estimated Delay Time.	High
FR-908	**Cost Per Mile (CPM) Analytics Tile:** The dashboard shall feature a real-time analytics tile displaying the current Fleet Cost Per Mile. The calculation shall be derived from the formula: `(Sum of Fuel Transactions + Sum of Maintenance Costs + Sum of Driver Payroll) / Total Miles Driven by Fleet`. The tile shall visually indicate the percentage change (increase/decrease) compared to the previous week/month.	High

-----3.8 Driver Management Module (Admin)

ID	Requirement Description	Priority
FR-1001	The Admin shall be able to view all driver details, including license number and years of experience.	High
FR-1002	The Admin shall be able to update driver information and reset their password.	High

FR-1003	The Admin shall be able to view a driver's historical shipment and emergency reporting records.	High
FR-1004	The Admin shall be able to suspend or activate a driver's account.	High

-----3.9 Truck Management Module (Admin)

ID	Requirement Description	Priority
FR-1101	The Admin shall be able to add, modify, and view truck details (Number, Type, Capacity, Status).	High
FR-1102	The system shall validate that Truck Number remains unique.	High
FR-1103	The Admin shall be able to assign a truck to a driver (or unassign).	High
FR-1104	The Admin shall be able to update truck status to "Maintenance" which removes it from available pool.	High
FR-1105	**Truck Profitability Analysis:** The system shall provide a detailed profitability report for each individual truck asset. This report shall calculate and display: `Total Shipment Revenue Generated by Truck` MINUS `(Fuel Costs Attributed to Truck + Maintenance Costs Attributed to Truck)`. The Admin shall be able to filter this report by date range.	High

3.10 Fuel Emergency and Payment Module

ID	Requirement Description	Priority
FR-1201	The driver shall be able to submit a fuel request including current location and required amount.	High
FR-1202	The Admin shall be able to approve the request and initiate payment via Razorpay API.	High
FR-1203	The system shall notify the driver and update the fuel request status to "Paid" upon successful payment.	High
FR-1204	The system shall handle payment failures and notify the Admin for manual resolution.	High
FR-1205	The system shall maintain a record of all fuel transactions with payment IDs.	Medium

-----4. Interface Requirements4.1 User Interfaces

ID	Requirement Description	Priority
IR-101	All interfaces (Customer, Driver, Admin) shall be responsive for optimal mobile and desktop use.	High
IR-102	The Driver interface shall prioritize large, distinct buttons for all emergency functions.	High
IR-103	The Customer interface shall prominently display the current shipment status and driver location map.	High
IR-104	The Admin dashboard shall use a modern, material design aesthetic.	Medium
IR-105	Error and notification messages shall be contextually relevant and non-intrusive.	Medium

4.2 Hardware Interfaces

ID	Requirement Description	Priority
IR-201	The Driver interface requires access to the device's GPS hardware for location tracking.	High
IR-202	The system requires microphone and speaker access on user devices for WebRTC calling.	High

4.3 Software Interfaces

ID	Requirement Description	Priority
IR-301	The system shall integrate with Twilio API for sending SMS and Email OTPs.	High
IR-302	The system shall use WebRTC protocol for in-app internet calling.	High
IR-303	The system shall use Socket.IO (or similar) for real-time chat functionality.	High
IR-304	The backend shall communicate with Google Maps/OpenStreetMap API for location services.	High
IR-305	All APIs shall be RESTful and use secure, authenticated endpoints.	High

4.4 Communication Interfaces

ID	Requirement Description	Priority
IR-401	All data communication, including WebSockets, shall be secured using the WSS (WebSocket Secure)	High

	protocol.	
IR-402	The system shall implement robust network error handling for intermittent internet connectivity.	Medium

-----5. Performance Requirements5.1 Response Time

ID	Requirement Description
PR-101	Login and key form submissions (e.g., shipment creation) shall complete within 3 seconds.
PR-102	Real-time chat messages shall be delivered within 500 milliseconds.
PR-103	Emergency alert notifications to Admin shall be delivered within 2 seconds of submission.
PR-104	Map loading and location display updates shall render within 4 seconds.

5.2 Capacity

ID	Requirement Description
PR-201	The system shall support up to 500 concurrent users (Admin, Driver, Customer).
PR-202	The system shall support efficient handling of up to 50,000 active customer accounts.
PR-203	The system shall support up to 5,000 concurrent WebSocket connections (for chat/location).

5.3 Availability

ID	Requirement Description
PR-301	The system shall be available 24/7 with a target uptime of 99.8%.
PR-302	Scheduled maintenance downtime shall be minimal and announced at least 48 hours in advance.

5.4 Throughput

ID	Requirement Description
PR-401	The system shall process at least 200 location updates per second during peak hours.
PR-402	The system shall process at least 15 payment transactions per minute.

-----6. Design Constraints6.1 Technology Stack

ID	Requirement Description
DC-101	The primary backend language shall be Python (Flask/Django) or Node.js.
DC-102	The front-end shall use a modern JavaScript framework (React/Vue/Angular).
DC-103	The database management system shall be MySQL or PostgreSQL.
DC-104	Communication protocols must include HTTPS, WSS, and WebRTC.

6.2 Security Constraints

ID	Requirement Description
DC-201	All user passwords and sensitive PII (Phone, License) shall be encrypted both in transit and at rest.
DC-202	Role-based access shall strictly control visibility of Customer and Driver PII (e.g., Driver phone number visibility).
DC-203	The system shall implement input sanitization to prevent XSS and SQL injection attacks.
DC-204	All third-party API keys (Twilio, Razorpay) shall be securely stored and rotated regularly.

-----7. Non-Functional Attributes7.1 Security

ID	Requirement Description
NFR-101	Multi-factor authentication (via OTP) shall be used during initial driver/customer registration.
NFR-102	Audit logs shall capture all communication, location, and emergency reporting activities.
NFR-103	Data transmission over public networks shall be secured via TLS/SSL.

7.2 Reliability

ID	Requirement Description
NFR-201	The system shall automatically handle and queue failed location updates for later synchronization.
NFR-202	Payment transaction state must be recoverable and idempotent.

NFR-203	The system shall use load balancing and redundancy to maintain uptime during peak loads.

7.3 Usability

ID	Requirement Description
NFR-301	The driver interface shall be simple and usable with minimal distraction while driving.
NFR-302	Customer tracking map shall be easily navigable and accessible via a single click from the dashboard.
NFR-303	All error and validation messages shall guide the user toward corrective action.

7.4 Scalability

ID	Requirement Description
NFR-401	The system architecture must be horizontally scalable to accommodate growth in truck and customer volume.
NFR-402	Database design shall be optimized for high-volume read operations (location tracking, dashboard display).

7.5 Maintainability

ID	Requirement Description
NFR-501	The codebase must be modular and documented according to established standards.
NFR-502	Automated deployment pipelines (CI/CD) should be established for faster releases.

-----8. Appendices8.1 Database Schema

*(Note: Schema for V2.0 would need to include new tables/fields for Customer, OTPs, Location tracking, and Chat logs.)*9.2 Glossary of Terms

Term	Definition
Customer	User initiating the shipment request.
Driver	User responsible for transport and location updates.
Admin	Transport company office user managing the system.
OTP	One-Time Password for delivery

	verification.
WebRTC	Web Real-Time Communication protocol for in-app calling.
Twilio	Third-party API service used for SMS and email delivery.
Shipment	Goods being transported from pickup to delivery.

-----9. Conclusion

The **FleetGuard – Smart Transport Management System Version 2.0** represents a significant upgrade, transforming the platform into a comprehensive logistics solution by directly connecting Customers, Drivers, and Admin. The integration of advanced features such as real-time location tracking, secure OTP verification for delivery, and multi-channel communication (WebRTC calling, chat) addresses modern logistics needs for visibility, security, and responsiveness.

This revised SRS outlines:

- **Ten enhanced functional modules** including new customer-facing and real-time features.
- **Comprehensive requirements** for three user roles (Customer, Driver, Admin).
- **Clear constraints** on performance, security, and third-party API dependencies.
- **A structured schedule and budget** for the increased complexity of V2.0.

The commitment to using technologies like Twilio and WebRTC, alongside robust security and performance planning, ensures that FleetGuard V2.0 will be a highly competitive, reliable, and user-centric system that minimizes communication gaps and maximizes delivery transparency

.-----*End of Software Requirements Specification Version*

2.0 Document